INSTITUTE OF PHYSICS PUBLISHING REPORTS ON PROGRESS IN PHYSICS

Rep. Prog. Phys. **64** (2001) 1815–1942 PII: S0034-4885(01)09455-6

Applications of B-splines in atomic and molecular physics

H Bachau[1], E Cormier[1], P Decleva[2], J E Hansen[3] and F Martín[4]

[1] Centre Lasers Intenses et Applications, UMR 5107 du CNRS, Université de Bordeaux I, 351 Cours de la Libération, F-33405 Talence, France
[2] Dipartimento di Scienze Chimiche, Via Giorgieri 1, I-34127 Trieste, Italy
[3] Department of Physics and Astronomy, University of Amsterdam, Valckenierstraat 65, NL-1018XE Amsterdam, The Netherlands
[4] Departamento de Química, C-9, Universidad Autónoma de Madrid, E-28049 Madrid, Spain

Received 6 July 2001
Published 19 November 2001
Online at stacks.iop.org/RoPP/64/1815

Abstract

One of the most significant developments in computational atomic and molecular physics in recent years has been the introduction of B-spline basis sets in calculations of atomic and molecular structure and dynamics. B-splines were introduced in applied mathematics more than 50 years ago, but it has been in the 1990s, with the advent of powerful computers, that the number of applications has grown exponentially. In this review we present the main properties of B-splines and discuss why they are useful to solve different problems in atomic and molecular physics. We provide an extensive reference list of theoretical works that have made use of B-spline basis sets up to 2000. Among these, we have focused on those applications that have led to the discovery of new interesting phenomena and pointed out the reasons behind the success of the approach.

 © 2001 IOP Publishing Ltd Printed in the UK

Contents

	Page
1. Introduction	1817
2. Description of B-splines	1820
2.1. The basis of B-splines	1822
3. Solving the Schrödinger equation with B-splines	1825
3.1. One-electron atomic systems	1825
3.2. Bound states	1827
3.3. Continuum states	1829
4. Atomic structure	1833
4.1. Methods for bound and quasi-bound states	1834
4.2. Evaluation of continuum states	1842
4.3. Applications	1849
5. Atoms in strong fields	1871
5.1. Perturbative approach for multiphoton processes	1871
5.2. Nonperturbative time-dependent methods	1878
5.3. Ions and atoms in static fields	1890
6. Diatomic molecules	1893
6.1. General considerations: the Born–Oppenheimer approximation	1894
6.2. The electronic Schrödinger equation	1895
6.3. The vibrational Schrödinger equation	1898
6.4. Continuum states	1899
6.5. Applications	1904
7. Polyatomic molecules and clusters	1910
7.1. Methods	1910
7.2. Applications	1916
8. Conclusion and perspectives	1923
Acknowledgments	1925
Appendix	1925
A.1. Breakpoint sequences	1925
A.1.1. Linear sequence.	1926
A.1.2. Exponential sequence.	1926
A.1.3. Sinelike sequence.	1926
A.1.4. Linear–parabolic sequence.	1926
A.2. Density of states	1927
A.3. Two-electron integrals	1927
References	1931

1. Introduction

The use of basis sets to solve the Schrödinger equation has a long history in physics. It transforms the solution of a differential equation into an algebraic eigenvalue problem, and, particularly in conjunction with the introduction of electronic computers, this approach has become popular since linear algebra is one of the best developed branches of numerical computation. Today large matrices can be routinely diagonalized with high accuracy in reasonably short times. Many types of basis set have been tried in the past but we focus in this review on the use of B-splines which is a relatively new development in atomic and molecular physics. In molecular physics, the use of basis sets of different types has been standard practice for a long time, but in atomic physics their use has been much more restricted. The reason is that the finite-difference methods, introduced by Hartree, for atoms with spherical symmetry have been found to be more accurate than the use of a limited basis set. This approach, in which a numerical wavefunction is determined at a limited number of mesh points in the self-consistent field (SCF) approach, using for example the Numerov algorithm, is well suited to atomic problems because of their spherical symmetry, which allows the separation of angular and radial variables. In particular, the multi-configuration Hartree–Fock (HF) and Dirac–Fock (DF) programs due to Froese Fischer and co-workers [337] and Grant and co-workers [338], respectively, have been found to be able to provide very accurate wavefunctions for atomic states. Generally speaking the reason for the superiority of the finite-difference methods for atoms is that the basis sets which have been tried need to be large in order to be 'complete enough', but linear dependences usually effectively restrict the size of basis that can be used before the accuracy attainable with the finite-difference methods has been reached.

B-splines have the property of being 'complete enough' with a relatively small number of basis functions and, since linear dependences are negligible, even for a large basis, it is possible for such a basis set to compete with the finite-difference methods. This accounts for the recent swing towards the use of B-splines in atomic calculations, while also for molecular problems B-splines have considerable advantages over earlier basis sets.

The finite-difference and the basis set methods can be characterized as 'local' and 'global' methods, respectively. In local methods, the solution at one point depends on the solution in nearby points only. As a consequence some initial estimate is required to get the algorithm started and normalization must be performed afterwards, while in the global methods the solution at one point is linked to the solutions at all other points so that an initial estimate is unnecessary and normalization is implicit. These are important advantages but there are compensating disadvantages with the usual basis sets. For example, for Slater type orbital (STO) basis sets, which have been the most popular basis for atomic calculations, nonlinear optimization is required. This is a difficult process to control, which means, for example, that the STO basis sets of HF quality established a long time ago by Clementi and Roetti [336] for the ground states of a number of atoms and ions still are routinely used as initial estimates for excited states of atoms because of the large amount of optimization which would be required to start the procedure from scratch. A consequence is that a good working knowledge of which basis set to use under particular circumstances is required. The use of a B-spline basis would not require nonlinear optimization, which is an important difference that still has to be exploited.

Splines of various types have been used, particularly for fitting purposes, in numerical analysis for a long time, and they are a standard part of fitting routines in commercial program packages. The name is derived from a draftsman's spline, a flexible piece of rubber, which can be used to draw smooth curves. The spline is pinned down at known points (called knot points

in the following) on the otherwise perhaps unknown or perhaps irregular curve. The flexibility of the spline now allows us to draw a continuous curve between the points with continuous first and second derivatives at all points. The use of splines in numerical analysis is designed to obtain the same properties in numerical fitting to known or partially known functions. Spline basis functions are compact L^2 integrable functions defined in a restricted space usually referred to as a box, in contrast to such L^2 basis sets as Gaussians or Slater orbitals that extend to infinity. In the box, a knot sequence is defined and the freedom to define the knots to suit the particular problem is, as we shall see, another important advantage of the method. The knot sequence defines the extent of the individual splines. Cubic splines, for example, are piecewise third-degree polynomials. They are used in many fitting programs and usually require a decision from the user concerning the boundary conditions to be used in each individual case. B-spline basis sets of order four are designed to provide the most compact basis for cubic splines and each spline is different from zero only between four adjacent knot points, except possibly the first and last splines. (Note that the order of the B-spline corresponds to the degree of the polynomials involved plus one. Both notations will be used throughout this review.) Thus B-splines are in principle defined as a set (B stands for basis) but for convenience we shall also refer to the individual splines in the set as B-splines. B-splines are positive definite functions, which leads to matrices that are easier to diagonalize. This is in contrast to cardinal splines, for example, that are different from zero over the whole box and not positive definite functions. Compactness has another advantage in matrix calculations, since integrals over two B-splines for example will be zero if the two splines are different from zero in disjoint intervals. The resulting matrices are sparse and often, but not always, banded, which is advantageous not only in the diagonalization procedure but also when calculating the matrix elements.

B-splines were introduced by Schoenberg in 1946 [1] and he also coined the name in 1967 [43], but it was only with the publication of the monograph by de Boor [5] that their application to atomic physics started in earnest. De Boor published FORTRAN routines that makes it possible to define and manipulate B-splines of arbitrary order and knot point distribution and this meant that the extension to B-splines of order higher than four became straightforward. This is essential for obtaining a competitive degree of accuracy in the solution of the Schrödinger equation. In the 1970s Shore published a series of papers [8–12] which nicely illustrates the development. In his first papers, Shore used cubic splines to solve a single-particle Schrödinger equation while in later work Shore apparently became the first to use B-splines for atomic structure calculations using mainly local potentials, and he introduced for example the exponential knot sequence to solve bound-state problems that we shall describe later. In his final paper Shore used B-splines of order higher than four but the main focus of his papers was to compare different basis sets, and he concluded that for comparable sizes B-splines provide the most accurate approximation to the true function, particularly for higher eigenvalues. Shore discussed exclusively different types of single-particle problem and did not consider SCF procedures.

Following Shore, Gilbert and co-workers [13, 14, 42] published work on the use of several types of spline as basis sets for SCF calculations. The first studies used cardinal splines but as mentioned above these have disadvantages in numerical calculations and therefore Altenberger-Siczec and Gilbert [42] switched to B-splines. The origin of their interest was molecular calculations, but the first HF application in 1976 was to He, and they concluded that the calculation of the HF operator in this basis using the so-called supermatrix approach required too much computer power. Alternatively the SCF procedure was proposed, in which the two-electron integrals should be recalculated for each iteration. Nevertheless, this was also deemed to be too time consuming to make a spline basis feasible. However, the advances in computer power have reduced this problem to manageable proportions. In addition, Altenberger-Siczec

and Gilbert based their conclusion on the use of cubic splines, and the introduction of higher-order splines means that basis sets smaller than was foreseen by them are required. The latter was in fact established shortly after the Altenberger-Siczes–Gilbert paper by Silverstone and collaborators [15–17]. Silverstone *et al* [16] also proposed that the use of piece-wise polynomials as basis would be an advantage for many-body perturbation theory (MBPT).

Nevertheless, it took another ten years with the attendant increase in computer power before the next attempt was made to make use of the properties of B-splines . First in 1986, on the instigation of Bottcher, Johnson and Sapirstein [104] introduced a B-spline basis set for the calculation of second- and higher-order corrections to properties of heavy atoms using MBPT. In their work, Johnson, Sapirstein and co-workers [99, 105] constructed orbitals in numerical form from the finite number of eigenfunctions to the DF equation for a valence electron in an alkaline atom in a B-spline basis. The set of numerical orbitals was used in the subsequent MBPT calculations. Thus the only property of the B-spline basis which is used is the 'completeness' that is retained when the wavefunctions are transformed to numerical form. This approach has the advantage that existing numerical computer programs can be used in the MBPT calculations with perhaps minor changes.

Other applications of B-splines take advantage of different properties of the basis. An approach that requires relatively simple computations is the collocation approach. However, this has been shown already by Shore to be much less accurate than the Raleigh–Ritz–Galerkin procedures which form the main subject of this review although the method has been revived by Bottcher and co-workers [28,29,32]. For Hermitian (symmetric) matrices the Raleigh–Ritz variational approach is well known, but for nonsymmetric matrices that appear, for example, when the boundary conditions are different from zero on the outer side of the box, the Galerkin approach is very useful, while for symmetric matrices it is equivalent to the Raleigh–Ritz procedure [339]. The various ways of implementing the Galerkin conditions is a prominent aspect of the technical part of this review.

Since the early work by Johnson and Sapirstein, B-splines have been used in many other types of calculation of various properties for both bound and continuum states and this is the main subject of the physics part of this review. In particular, B-splines are able to provide a very accurate representation of continuum states, which makes them superior to more conventional L^2 basis sets. Although the first applications were to static properties, it has turned out that B-splines perhaps are even more important for the calculation of dynamic properties and we shall discuss subjects such as multiphoton excitation, above-threshold ionization (ATI) and high-order harmonic generation (HOHG) in atoms and dissociation of molecules for which the use of B-splines have meant that effects could be taken into account which theoretically were beyond reach earlier.

The structure of the review is as follows. In the next two sections we first describe briefly the mathematical properties of B-splines and then how to use them to solve the single-particle equation. Section 4 describes the basic techniques for more electrons and applications. Readers who have little previous experience with B-splines would want to start with section 3 while readers with some knowledge of splines and interested in applications might go directly to section 4 (atomic structure), 5 (atoms in strong fields), 6 (diatomic molecules) or 7 (polyatomic molecules and clusters), depending on their particular interest. Sections 5–7 are fairly independent of each other. They are also independent of section 4.3, which is exclusively devoted to applications in atomic structure calculations. To reduce the size of the review we shall largely restrict ourselves to nonrelativistic basis sets and applications, except for relativistic configuration interaction (CI), which is close to other nonrelativistic applications.

We end with a bibliography of all publications we have found concerning the use of B-splines in atomic and molecular physics. The bibliography is very large and not all the

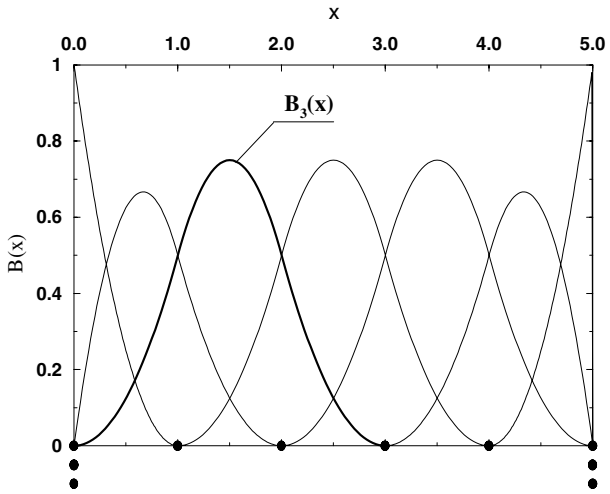


Figure 1. The full set of B-splines of order $k = 3$ relative to the knot sequence $\{0, 0, 0, 1, 2, 3, 4, 5, 5, 5\}$. Knots are represented by full circles.

references are discussed in the review. To help the reader we have divided the papers into sections although some papers in practice cover more than one category. The general references have been given separately for each section to reduce the time needed to find a particular reference. If a reference is mentioned in several chapters it is included only once (the first time).

Throughout, we use atomic units (au) unless specifically indicated.

2. Description of B-splines

B-splines are functions designed to generalize polynomials for the purpose of approximating arbitrary functions. One can view them as new elementary functions such as $\sin x$ or $j_l(x)$. One becomes familiar with them by understanding their qualitative behaviour and how to use them, that is how to obtain values of the functions, their derivatives or integrals. This is the topic of this section. A complete description of B-splines and their properties can be found in de Boor's book [5]. Just like $\sin x$, efficient numerical evaluation can be accomplished by black-box routines, and it is not generally important to know the algorithm for proficient use.

Let us introduce a few definitions.

- The polynomials of order k (maximum degree $k - 1$) are
$$p(x) = a_0 + a_1 x + \cdots + a_{k-1} x^{k-1}.$$
- A function which is continuous (on a given interval) together with its derivatives up to order n, that is $f(x)$, $Df, \ldots, D^n f$, is said to be of class C^n. Then C^0 means that only f is continuous and C^{-1} that f is discontinuous.
- Consider an interval $I = [a, b]$ divided into l subintervals $I_j = [\xi_j, \xi_{j+1}]$ by a sequence of $l + 1$ points $\{\xi_j\}$ in strict ascending order
$$a = \xi_1 < \xi_2 < \cdots < \xi_{l+1} = b.$$
The ξ_j will be called *breakpoints* (bps).

A full set of B-splines (to be defined later) over the interval $[0, 5]$ is shown in figure 1, which we shall repeatedly use for illustration. In this example, the interval is divided into five

subintervals by the bps $\{0, 1, 2, 3, 4, 5\}$. Each B-spline (consider B_3 for example) is a function made up of different polynomial pieces on adjacent subintervals, of a fixed order k ($k = 3$ in the present case), joined with a certain degree of continuity at the bps. In this case each B-spline is continuous together with its first derivative (class C^1) at the interior bps, while at $x = 0$ B_3 is still C^1, but B_2 has discontinuous first derivative, and B_1 is discontinuous (they are C^0 and C^{-1} respectively). The same is true for B_6 and B_7 at $x = 5$, and it will be typical of the B-spline bases commonly employed. An explicit representation of $B_3(x)$ in the example is

$$B_3(x) = \begin{cases} \frac{1}{2}x^2 & 0 \leqslant x < 1 \\ x^2 + 3x - \frac{3}{2} & 1 \leqslant x < 2 \\ \frac{1}{2}x^2 - 3x + \frac{9}{2} & 2 \leqslant x < 3 \\ 0 & \text{otherwise} \end{cases}$$

although this is rarely needed.

- Let us associate with interior bps ξ_j, $j = 2, \ldots, l$, a second sequence of non-negative integers ν_j, $j = 2, \ldots, l$, $\nu_j \geqslant 0$, which define continuity condition $C^{\nu_j - 1}$ at the associated bps ξ_j. With the end bps ξ_1 and ξ_{l+1} we associate $\nu_1 = \nu_{l+1} = 0$, that is we do not require any continuity. This is natural since we are only interested in the interval $[a, b]$. Further restrictions may be dictated by boundary conditions at the endpoints, and are easily implemented as we shall see. In the example we therefore have $\{\nu_j\} = \{0, 2, 2, 2, 2, 0\}$.
- Finally let us call *knots* another sequence of points $\{t_i\}$ in ascending order, not necessarily distinct,

  $$\{t_i\}_{i=1,\ldots,m}, \qquad t_1 \leqslant t_2 \leqslant \cdots \leqslant t_m$$

  associated with ξ_j and ν_j as follows:

  $$\begin{aligned} &t_1 = t_2 = \cdots = t_{\mu_1} = \xi_1; && \mu_1 = k \\ &t_{\mu_1+1} = \cdots = t_{\mu_1+\mu_2} = \xi_2 \\ &\cdots \\ &t_{p+1} = \cdots = t_{p+\mu_i} = \xi_i; && p = \mu_1 + \mu_2 + \cdots + \mu_{i-1} \\ &\cdots \\ &t_{n+1} = \cdots = t_{n+k} = \xi_{l+1}; && \mu_{l+1} = k; \qquad n = \mu_1 + \cdots + \mu_l \end{aligned}$$

  where μ_j is the multiplicity of the knots t_i at ξ_j and is given by $\mu_j = k - \nu_j$. In the example $\{t_i\} = \{0, 0, 0, 1, 2, 3, 4, 5, 5, 5\}$ and $\{\mu_j\} = \{3, 1, 1, 1, 1, 3\}$. Actually only multiplicity at the inner bps is important, and we shall always choose maximum multiplicity $\mu_1 = \mu_{l+1} = k$ at the endpoints.

The most common choice for knot multiplicity at inner bps is unity, corresponding to maximum continuity, that is C^{k-2}. This choice will be employed throughout this present paper, unless otherwise stated. With this choice the number of B-spline functions n ($n = 7$ in the example) is given by

$$n = l + k - 1.$$

Continuity may be relaxed at the inner bps by increasing the associated knot multiplicity there.

Any function expressed as a linear combination of the B_i will be called a *piecewise polynomial function* (pp-function) over $[a, b]$.

$$f = \sum_{i=1}^{n} c_i B_i$$

and it can be shown [5] that f is the most general function made of l polynomial pieces of order k, one for each subinterval I_j joined at the interior bps with continuity C^{ν_j-1}. Note that there is no point in asking for more than C^{k-2} continuity, since otherwise f reduces to a single polynomial over all I, so some discontinuity is the essence of pp-functions. On the other hand it is rare to employ discontinuity in the lowest derivatives or the function itself. If this is the case, for instance with no continuity constraints, $f \in C^{-1}$, one should be careful to consider that $f(\xi_j^-) \neq f(\xi_j^+)$; the function has different left and right limits at the bps. To make f and $D^r f$ uniquely defined over all I, de Boor [5] arbitrarily chooses right continuity, $f(\xi_j) = f(\xi_j^+)$. Careful consideration of discontinuities is, however, not necessary if they are confined to derivatives higher than those actually needed, which is always the case in applications. We define the rth derivative of a pp-function, $D^r f$, as the pp-function of order $k - r$ which has the same sequence ξ and is written in terms of $D^r p_j$ polynomials.

2.1. The basis of B-splines

A single B-spline $B(x)$ is defined by the order $k > 0$, and a set of $k + 1$ knots, $\{t_i, \ldots, t_{i+k}\}$, such that $t_i < t_{i+k}$.

Important properties, apparent in figure 1, are the following.

- $B(x)$ is a pp-function of order k over $[t_i, t_{i+k}]$.
- $B(x) > 0$ for $x \in]t_i, t_{i+k}[$.
- $B(x) = 0$ for $x \notin [t_i, t_{i+k}]$.
- For $x = \xi_j$ $B(x) \in C^{k-1-\mu_j}$, where μ_j is the multiplicity at ξ_j. The maximum multiplicity is k, giving discontinuous functions, the minimum is one, giving $B(x) \in C^{k-2}$.
- The knots need not be equidistant and the shape of $B(x)$ changes smoothly with the change of the knots.

In general a family of B-spline functions, $B_i(x)$, $i = 1, \ldots, n$ is completely defined given $k > 0$, $n > 0$ and a sequence of knots $\boldsymbol{t} = \{t_i\}_{i=1,\ldots,n+k}$ (actually $\{t_i\}$ may be infinite at one or both ends). Since k and t are usually fixed, one writes simply $B_{t,i}^k = B_i$. Each B_i is defined over an interval $[t_i, t_{i+k}]$, which contains $k + 1$ consecutive knots, and is indexed by the knot where it starts, so exactly one B_i starts at each knot t_i, $i = 1, \ldots, n$, and ends k knots later (see figure 1). In the example B_3 starts at $t_3 = 0$ and ends at $t_6 = 3$. $B_i > 0$ over the interval $]t_i, t_{i+k}[$, which is called the support of B_i, and $B_i = 0$ outside it. In fact, of all pp-functions, B-splines are designed to have minimal support. Some general properties of B-splines are the following.

- Over each interval $]t_i, t_{i+1}[$, $t_i < t_{i+1}$, exactly k B-splines are nonzero,

  $$B_j(x) \neq 0 \qquad \text{for} \quad j = i - k + 1, \ldots, i$$

  the first being B_{i-k+1}, which ends at t_{i+1}, and the last is B_i, which starts at t_i. Therefore we have identically

  $$B_i(x) \cdot B_j(x) = 0 \qquad \text{for} \quad |i - j| \geqslant k.$$

- In the expansion of an arbitrary function

  $$f(x) = \sum_{j=1}^{n} c_j B_j(x) = \sum_{j=i-k+1}^{i} c_j B_j(x) \qquad \text{for} \quad x \in [t_i, t_{i+1}]$$

  one always has only k terms contributing, so a minimal number of operations is needed.

- Because the B-splines are non-negative with minimal support, the expansion coefficients of an arbitrary function f are close to the function values at the knots. This means that wild oscillations in the coefficients are avoided, cancellation errors are minimal and numerical stability maximal (see for instance figure 4 in section 3.2).
- Each interval $I_j = [\xi_j, \xi_{j+1}] = [t_i, t_{i+1}]$ is characterized by a pair of consecutive knots $t_i < t_{i+1}$. t_i is called the left knot of interval I_j, and determines the indices of the B_i contributing over I_j, which are $B_{i-k+1}, \ldots, B_i$.
- They are normalized as $\sum_i B_i(x) = 1$ over the whole $[t_k, t_n]$.
- For simple equidistant knots each B_i is just a translation by one interval of the previous one. If the knots are not equidistant there is a smooth change in shape.
- B-splines [5, p 130], satisfy the recursion relation

$$B_i^k(x) = \frac{x - t_i}{t_{i+k-1} - t_i} B_i^{k-1}(x) + \frac{t_{i+k} - x}{t_{i+k} - t_{i+1}} B_{i+1}^{k-1}(x). \tag{1}$$

  Together with the definition of B-splines of order $k = 1$

$$B_i^1(x) = 1 \qquad t_i \leqslant x < t_{i+1} \qquad \text{and} \qquad B_i^1(x) = 0 \quad \text{otherwise}$$

  this gives rise to the algorithm employed for the practical evaluation of B-splines: given a point x, one generates by recursion the values of all the k B-splines which are nonzero at x, as illustrated in figure 2. The derivative of a B-spline of order k, being a pp-function of order $k - 1$, can be also expressed as a linear combination of B-splines of the same order [5, p 138],

$$DB_i^k(x) = \frac{k-1}{t_{i+k-1} - t_i} B_i^{k-1}(x) - \frac{k-1}{t_{i+k} - t_{i+1}} B_{i+1}^{k-1}(x).$$

Actually there is complete freedom in the choice of the first and last k knots, the only requirement being

$$t_1 \leqslant \cdots \leqslant t_k \leqslant \xi_1 \qquad \xi_{l+1} \leqslant t_{n+1} \leqslant \cdots \leqslant t_{n+k}.$$

Their choice affects the behaviour of the first and last k B-splines, but is of no consequence on their basis set property for pp-functions on $[\xi_1, \xi_{l+1}]$. The standard choice $t_1 = \cdots = t_k = \xi_1$ and $t_{n+1} = \cdots = t_{n+k} = \xi_{l+1}$ is particularly convenient. In this case (see figure 1) all $B_i = 0$ outside $[a, b]$. Moreover only the first B-spline is discontinuous at $x = a$, $B_1(x) \neq 0$, the same for DB_1 and DB_2, and likewise for the higher derivatives; the same at $x = b$ for B_n, DB_n and DB_{n-1}, and so on. This makes it very easy to implement boundary conditions at the endpoints. For instance $f(a) = 0$ is satisfied by deleting B_1 from the set, $Df(b) = 0$ by combining B_{n-1} and B_n as $\tilde{B}_{n-1} = c_1 B_{n-1} + c_2 B_n$ so that $D\tilde{B}_{n-1}(b) = 0$, and so on.

The most common choice is therefore to employ knots of multiplicity k at the endpoints and simple knots at the interior points, unless there is a known singularity at some point, where continuity may be reduced by suitably increasing the knot multiplicity. In general, when approximating analytic functions, it is best to employ high-order splines, compatible with the numerical stability and roundoff errors, typically in the range $k = 7$–10. Note that the error will be bound by

$$\varepsilon \sim \frac{h_j^k}{k!} |D^k f(\eta_j)|$$

where h_j is the width of the interval I_j and $\eta_j \in I_j$. This is the main advantage of B-splines over global bases, in that the error can be controlled by step size, in an analogous manner to finite-difference approaches, but retaining all the advantages of basis set expansions. The completeness property of the basis may be characterized by the fact that for a given threshold

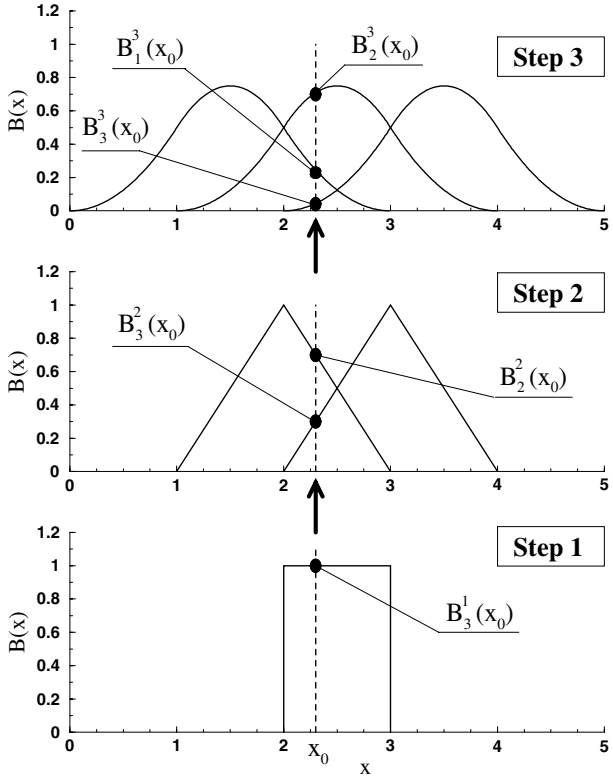


Figure 2. Recursive evaluation of B-splines up to order $k = 3$, relative to the knot sequence $\{0, 1, 2, 3, 4, 5\}$, which are not zero at $2 < x < 3$. Each set is obtained from the previous one by applying the recursion formula (1). Notice that at each x one obtains k nonzero B-spline values, which add up to unity.

ε all sufficiently smooth functions will be approximated within ε, provided that the derivatives are conveniently limited, that is up to a cut-off on the maximum local value of the derivatives. We shall see in section 3.3 that, in the quantum mechanical case, this translates into a limit on the maximum kinetic energy of the states considered, which can be tuned by the choice of the interval step h.

From a practical point of view, stable evaluation of splines is accomplished by the recursion algorithm (1) (see chapter X in [5]) and is efficiently implemented in subroutine BSPLVP (ibid., p 134), which requires as input values the spline order, the knot sequence, the value of the abscissa and the index of the 'left knot' t_i. The latter can be located given the abscissa and the knot sequence, by subroutine INTERV (ibid., p 92). The routine returns the values of the k B-splines that are nonzero at x, that is $B_{i-k+1}(x), \ldots, B_i(x)$. If derivatives are also needed they can be evaluated as combination of B-splines of lower order by subroutine BSPLVD (ibid., p 288).

In quantum mechanical applications mostly matrix elements, i.e. definite integrals involving B-splines and their derivatives, are involved. These are best computed to machine accuracy employing Gauss–Legendre (GL) integration [6, 7] of appropriate order over each subinterval I_j. Recall that GL of order n (n points) is exact for a polynomial integrand of order $2n$. So k GL points integrate exactly the product $B_i B_j$, and a little more is usually completely

accurate even for nonpolynomial integrands, provided no singularity is present in the interior of I_i (see appendix A.3).

3. Solving the Schrödinger equation with B-splines

This section introduces the basic techniques in the description of atomic structure with B-spline basis set. Before extending those techniques to multi-electron atoms and even more complicated systems such as molecules, we focus on the simplest atomic systems: those with one electron bound in a central potential. This class of systems obviously includes the hydrogen atom and any hydrogen-like ions characterized by the Coulomb potential, but also true one-valence electron systems such as alkaline atoms (Na, K, Rb, Cs, ...) or alkaline-like ions (Be^+, Mg^+, ...) provided a sufficiently accurate model or pseudo-potential is available to describe the frozen ion core potential. Less intuitive is the modelling of closed-shell systems such as rare gases in the single-active-electron (SAE) approximation [340, 341]. For example, a model potential recently appeared in the literature [342] that is able to describe the argon atomic structure and fairly well reproduce the dynamics of ATI by intense laser fields.

As demonstrated below, the flexibility of the present approach allows us to calculate both the energies and the wavefunctions in a given central potential with an arbitrary accuracy that can be as high as the machine precision on table top workstations taking only a few minutes. The limitations of the potential will determine the overall accuracy.

3.1. One-electron atomic systems

An eigenstate $\psi(\boldsymbol{r})$ with energy E describing a single-electron atom is a solution of the stationary Schrödinger equation:

$$[-\tfrac{1}{2}\boldsymbol{\nabla}^2 + V(\boldsymbol{r})]\psi(\boldsymbol{r}) = E\psi(\boldsymbol{r}) \tag{2}$$

where $V(\boldsymbol{r}) = V(r)$ is the atomic central potential. Accounting for the symmetry, it is more convenient to separate the spherical variables and to look for solutions of the form

$$\psi_{nlm}(\boldsymbol{r}) = \frac{\mathcal{U}_{nl}(r)}{r} Y_l^m(\theta, \phi) \tag{3}$$

where $Y_l^m(\theta, \phi)$ is a spherical harmonic function. The radial function $\mathcal{U}_{nl}(r)$ is thus a solution for the reduced Schrödinger equation:

$$\left[-\frac{\mathrm{d}^2}{2\,\mathrm{d}r^2} + \frac{l(l+1)}{2r^2} + V(r)\right]\mathcal{U}_{n,l}(r) = E_{n,l}\mathcal{U}_{n,l}(r) \tag{4}$$

with the boundary condition $\mathcal{U}_{n,l}(r = 0) = 0$.

The radial equation (4) is solved numerically in a subspace by assuming that any solution $\mathcal{U}_{nl}(r)$ can be approximated, in a given cavity, by spline functions. As explained in section 2.1, such a subspace can be generated by the basis of B-spline functions. It is thus natural to expand the solutions belonging to that space onto the B-spline basis set as

$$\mathcal{U}_{nl}(r) = \sum_{i=1}^{N} c_i^{nl} B_i(r) \tag{5}$$

where $B_i(r)$ is the ith B-spline of order k as defined in section 2.1. Building the basis requires a knot sequence (see section 2.1 for a description of the full procedure) which depends on the following parameters: a set of mesh points called the breakpoint sequence defined in $[0, r_{\max}]$ (see appendix A.1 for a list of possible arrangements), the order k of the piecewise polynomials

and the continuity conditions at each breakpoint. All these parameters are chosen the properties of the physical problem in mind.

In the present case, the cavity (often called the box), in which we solve the problem, is defined by letting the radial coordinate r vary within the interval $[0, r_{\max}]$. This interval is now divided into segments whose endpoints form the breakpoint sequence. There is a total freedom in choosing the mesh of bps. An optimal choice will depend on the type of result we are particularly interested in. For instance, to obtain an accurate bound-state wavefunction, it is more appropriate if the bps are concentrated close to the nucleus (exponential, polynomial or power breakpoint sequence) because the solution vanishes at some finite distance. In contrast, a continuum wavefunction, which oscillates to infinity, is therefore uniformly represented if the bps are equally spaced (linear sequence). Alternatively, one may need accuracy for the initial ground state (GS) and, at the same time, a fairly good description of the continuum as is the case in ionization cross-section calculations or in the study of ATI for example (see section 5.2). These applications use a mixed sequence which is polynomial up to some distances from the nucleus and linear afterwards. There is no physical reason for not imposing the maximum-continuity conditions ($\nu_i = k - 1$) at each breakpoint in $]0, r_{\max}[$. However, to satisfy the boundary conditions, we require the solution to vanish at the endpoints: $\mathcal{U}_{n,l}(r = 0) = \mathcal{U}_{n,l}(r = r_{\max}) = 0$ (see section 3.3 for a different choice of boundary conditions at $r_{\max}$). These conditions are fulfilled either by removing the first and last B-splines from the basis set or by setting the continuity conditions $\nu_i = 1$ at the boundaries.

For a given fixed angular momentum l, the atomic states (energy and wavefunction) satisfying (4) are calculated by solving the system of N linear equations obtained when substituting (5) in (4) and projecting on $B_j(r)$. Written in matrix form, this procedure is equivalent to solving the following generalized eigensystem:

$$\boldsymbol{H}_l \cdot \boldsymbol{c} = E\ \boldsymbol{S} \cdot \boldsymbol{c} \tag{6}$$

for E_{nl} and $\boldsymbol{c}_{nl} = \{c_i^{nl}\}_1^N$, where

$$\begin{aligned}(\boldsymbol{H}_l)_{ij} = &-\frac{1}{2}\int_0^{r_{\max}} B_i(r)\frac{\mathrm{d}^2}{\mathrm{d}r^2}B_j(r)\,\mathrm{d}r + \frac{l(l+1)}{2}\int_0^{r_{\max}}\frac{B_i(r)B_j(r)}{r^2}\,\mathrm{d}r \\ &+ \int_0^{r_{\max}} B_i(r)V(r)B_j(r)\,\mathrm{d}r \\ (\boldsymbol{S})_{ij} = &\int_0^{r_{\max}} B_i(r)B_j(r)\,\mathrm{d}r.\end{aligned} \tag{7}$$

The overlap matrix $\boldsymbol{S}$ originates from the fact that B-splines do not form an orthonormal set of basis functions. All matrix elements are computed exactly using a Gauss quadrature. A GL quadrature with n GL points calculates exactly the integral of a polynomial of degree $2n + 1$ over a closed interval. Since a B-spline of order k is a particular polynomial of degree $k - 1$ on each segment, one easily sees that the integrands of the element $(\boldsymbol{S})_{ij}$ and the kinetic term of the element $(\boldsymbol{H}_l)_{ij}$ are polynomials with maximum degree $2k - 2$. Applying the GL procedure *on each segment* leads to an exact numerical evaluation of these two integrals. The case of the centrifugal and the potential term of the element $(\boldsymbol{H}_l)_{ij}$ is slightly different. Here the integrand is no longer a polynomial of some order but rather a rational fraction and the above statement does not hold. However, it can be shown that a small increment in the number of GL points results in an immediate convergence to machine accuracy. All matrix elements are therefore implicitly computed exactly.

Another key property directly related to the use of a B-spline basis follows from the fact that B-splines are compact functions: a B-spline of order k differs from zero only on k successive segments. Therefore, any local operator expressed in the B-spline basis appears

Table 1. Energies (in au) of all eigenstates of hydrogen (s states) computed with B-splines and compared with the exact values ($k = 9$, $r_{\max} = 100$ au, $N = 100$, exponential sequence with $\gamma = 5$).

n	E_n	$E_n^{\text{exact}} = -1/2n^2$	ΔE
1	−0.500 000 0000	−0.500 000 0000	9.0×10^{-13}
2	−0.125 000 0000	−0.125 000 0000	1.0×10^{-13}
3	−0.055 555 5555	−0.055 555 5555	4.4×10^{-14}
4	−0.031 249 9999	−0.031 250 0000	3.9×10^{-13}
5	−0.019 999 9715	−0.020 000 0000	2.8×10^{-08}
6	−0.013 868 4777	−0.013 888 8889	2.0×10^{-05}
7	−0.009 596 3749	−0.010 204 0816	6.1×10^{-04}
8	−0.004 662 8369	−0.007 812 5000	3.1×10^{-03}

as a matrix having nonvanishing values on a diagonal band of width $2k - 1$. Moreover, the Hamiltonian as well as the overlap matrix are symmetric matrices. It thus turns out that, for a given l, out of the N^2 matrix elements, only Nk need to be computed and stored in memory.

Solving the eigensystem (6) is now straightforward if one accounts for the fact that the overlap matrix $\boldsymbol{S}$ is positive definite. The symmetric band generalized eigensystem (6) can then be transformed into a regular eigensystem with the same band structure. A direct diagonalization of the new system provides all eigenenergies but this procedure is not able to recover the corresponding eigenfunctions since the transform matrix (full N^2 matrix) has been dropped. Having all eigenvalues, the eigenvectors are computed one by one in a second step by inverse iteration when necessary. This procedure never requires more than three iterations for a relative error of the order of the machine accuracy.

3.2. Bound states

The diagonalization procedure provides a set of eigenenergies that are associated with two different kinds of state. The energies lying below the ionization threshold (negative energies in the case of a central potential vanishing at infinity) correspond to the bound atomic states while the energies above that threshold are associated with continuum states. This subsection and the next one are devoted to explaining the influence of the basis parameters on the atomic bound states as well as to giving an interpretation of the so-called discretized continuum states. In what follows, the discussions are illustrated in the context of atomic hydrogen.

An example of a bound-state calculation is given in table 1, where we show all bound states (with angular momentum $l = 0$) obtained with the following B-spline basis: $k = 9$, $r_{\max} = 100$ au, $N = 100$, exponential breakpoint sequence with $\gamma = 5$ (see appendix A.1 for definitions of breakpoint sequences). Although the basis size is rather small, the agreement is excellent (machine accuracy) for the first four states. This is partly due to the choice of an exponential breakpoint sequence. Indeed, bound states vary close to the nucleus before vanishing exponentially at larger distances. Such a sequence defining a finer mesh (small segments) at small distances and larger segments away from the nucleus is therefore better adapted to the computation of bound states [9, 322]. As an example, the GS energy of hydrogen computed with the same basis but a linear sequence gives a deviation from the exact value of $\Delta E = 2.4 \times 10^{-4}$ au. Alternatively, one can use other types of sequence that concentrate bps near the nucleus such as power, sinelike, polynomial etc sequences (see appendix A.1).

Table 2. The different sets of parameters (k, N) achieving a particular level of accuracy ΔE on the GS of hydrogen. The optimal set (depicted with bold-face type) is chosen as being the least CPU time consuming (linear sequence with $r_{\max} = 1000$ au).

	$\Delta E < 10^{-3}$		$\Delta E < 10^{-6}$		$\Delta E < 10^{-12}$	
k	N	Time	N	Time	N	Time
4	860	17	3000	174.5		
5	530	10	1370	49.1		
6	**370**	**9.7**	860	31.7	4000	403.5
7	280	12.1	**620**	**31.6**	2370	243
8	220	16.1	490	39.5	1590	192
9	190	23.4	400	52.1	**1160**	**175**
10			340	70	890	201
11					710	242
12					620	316

In our context, the B-spline approach requires that the problem is confined in a sphere, which is in contradiction with the fact that, for a Coulomb potential, the spatial extension of the wavefunction tends to infinity as the principal quantum number increases. In fact, in hydrogen, one has $\langle r\rangle_{nl} = \frac{1}{2}\left[3n^2 - l(l+1)\right]$. Obviously, the box size $r_{\max}$ will limit the number of bound states obtained by this technique. As an example, although $\langle r\rangle_{1s} = 1.5$ au, it requires a box of $r_{\max} = 30$ au to obtain the proper energy of E_{1s}. Therefore, $r_{\max}$ must be set large enough for the box to contain all the atomic bound states of interest. The increasing deviation ΔE, shown in table 1, as n gets bigger is not a consequence of inaccuracy but is due to the fact that high-n states are eigenstates of the atomic Hamiltonian within the box (often referred to as pseudo-states). Indeed, the boundary conditions imposed here are equivalent to adding, to the atomic potential, a potential step of infinite height located at $r = r_{\max}$.

Given a box size $r_{\max}$ and a breakpoint sequence, higher accuracy is achieved by either increasing the order k or the density of bps, that is, since $r_{\max}$ is fixed, the basis dimension N. The rate of convergence is different for the two parameters as well as the increase in computational cost. Thus, depending on the degree of accuracy required, there exist optimal sets of parameters (k, N) in terms of CPU time as shown in table 2. This table clearly shows why high-order splines are much more efficient than simple cubic splines ($k = 4$). In practice, the order k is chosen between six and nine (the range commonly used in the literature being $k \in [7, 11]$) and the basis dimension adjusted to meet the desired accuracy.

The accuracy and computational stability of the B-spline approach compared with other basis expansions is clearly illustrated by comparing figures 3 and 4. Both represent a plot of the 4s wavefunction in H but one is built with STOs while the other uses B-splines. Figure 3 shows that the value of the wavefunction results from large cancellations in the linear combination of STOs (things worsen as n is increased). Cancellations, on the other hand, are dramatically minimized when B-splines are used, as seen in figure 4, due to the property that B-splines have minimal support. Indeed, stability in the wavefunction evaluation is enforced by the fact that there exist only k nonvanishing B-splines of order k at any point. The result is that the sum (see equation (5)) involved in the evaluation of the wavefunction at a particular position reduces to

$$\mathcal{U}_{nl}(r) = \sum_{j=i-k+1}^{i} c_j^{nl} B_j(r) \tag{8}$$

where $r \in [t_i, t_{i+1}]$. The evaluation thus always involves the sum of k terms whatever the basis dimension N is.

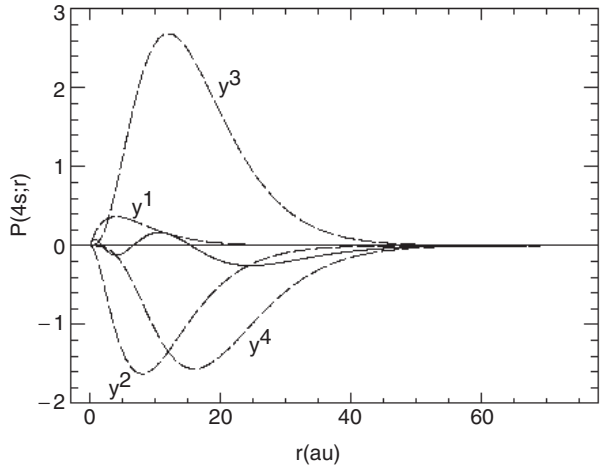


Figure 3. Expansion of the 4s wavefunction in H (solid curve) in terms of four STOs. The dashed curves represent the contribution of each orbital ($y^n = A_n r^n \mathrm{e}^{-\alpha r}$) to the wavefunction. Note the large cancellations between the various components (from [321]).

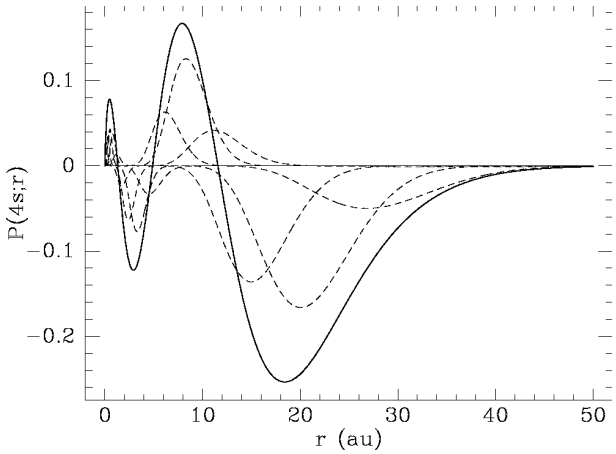


Figure 4. Expansion of the 4s wavefunction in H in terms of B-splines. The dashed curves represent the contribution of each B-spline i of order $k = 5$ (the terms $c_i B_i(r)$ in equation (5)). Note here the quasi-absence of cancellations (from [321]).

3.3. Continuum states

Formally, the solutions of the Schrödinger equation for positive energies appear as a family of functions depending on the continuous parameter E. In our approach however, the diagonalization procedure only provides a discrete set of electronic wavefunctions. These are the particular stationary solutions satisfying $\mathcal{U}_{nl}(r = r_{\max}) = 0$ due to the fact that we have restricted the problem to a finite part of the space. Nevertheless, these functions can be interpreted as a representation of the true electronic continuum with a different normalization (see below). One of the great advantages of making use of B-spline functions is that the density of discretized continuum states can be adjusted with a total degree of freedom simply by modifying $r_{\max}$ as shown in figure 5.

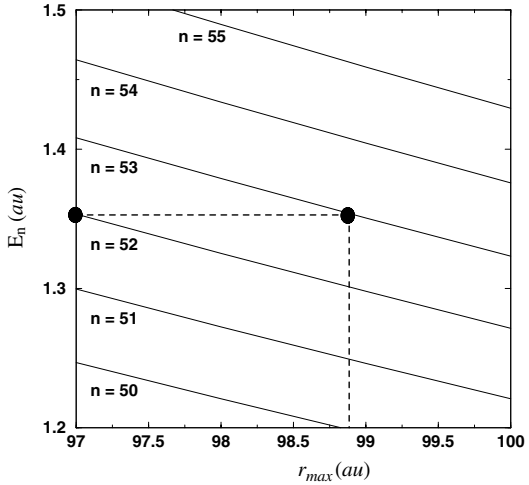


Figure 5. Plot of a selection of discretized continuum energies as a function of $r_{\max}$ ($N = 100$). Note that by varying $r_{\max}$ from 97 to 98.9 au, state number 53 will scan the empty energy gap that separates it from state number 52. Note also that the curves are not straight lines but vary as $1/r_{\max}^2$.

The box size $r_{\max}$ is a very important and useful parameter for manipulating discretized continuum states. The value of $r_{\max}$ uniquely sets the energy of each discretized continuum state and, consequently, the density of states. This is better understood if one considers the asymptotic behaviour of a Coulomb partial continuum wavefunction, which reads, when $kr \to \infty$,

$$\mathcal{U}_{E,l}(r) = \sqrt{\frac{2}{\pi k}} \sin\left(kr + \frac{Z}{k}\ln(2kr) - \frac{l\pi}{2} + \arg\,\Gamma\left(l + 1 - \frac{\mathrm{i}Z}{k}\right) + \sigma_l\right) \quad (9)$$

where $k = \sqrt{2E}$, l is the angular momentum, Z the nuclear charge, $\arg\,\Gamma(l + 1 - \mathrm{i}Z/k)$ the Coulomb phase and σ_l the non-coulombic phase ($\sigma_l = 0$ for hydrogen). Since our method implies that all solutions of the Schrödinger equation vanish at $r = r_{\max}$ ($\mathcal{U}_{E,l}(r_{\max}) = 0$), a discretized wavefunction at energy $E = k^2/2$ satisfies

$$kr_{\max} + \frac{Z}{k}\ln(2kr_{\max}) - \frac{l\pi}{2} + \arg\,\Gamma\left(l + 1 - \frac{\mathrm{i}Z}{k}\right) + \sigma_l = n\pi \quad (10)$$

where n is an integer. Equation (10) defines a relation between the energy and a discrete parameter n, which can be used as a wavefunction index. In a first approximation (when $kr_{\max}$ is bigger than all the other terms) this relation leads to the energy of a particle in a box [330]:

$$E_n = \frac{n^2\pi^2}{2r_{\max}^2}. \quad (11)$$

This latter property, which is illustrated in figure 5, makes it possible to compute continuum wavefunctions at any energy E simply by choosing the proper value of $r_{\max}$. This is achieved either by solving equation (10) for $r_{\max}$ with $k = \sqrt{2E}$, or through an inverse iteration procedure [343] which varies the box size until $E_n = E$. Some applications, like the computation of multiphoton ionization (MPI) cross-sections, may require a resolution in the final-state energy sometimes smaller than the energy spacing obtained through the discretization procedure. There are two ways to tackle this problem. One of them is to repeat

the computation with slightly different box sizes. For example, as shown in figure 5, varying $r_{\max}$ between 97 and 98.9 au shifts the energy of state $n = 53$ from 1.41 down to 1.35 au. Notice that this trick barely affects the energy spacing. The other way is to increase the box size (keeping the breakpoint spacing constant), thus reducing the energy spacing. As a matter of fact, the quality of the continuum discretization improves as the density of states increases (i.e. the energy spacing decreases). The density, defined as $\rho(E_i) = \frac{\partial n}{\partial E_n}|_{n=i}$ (see below), can be evaluated from equation (11) as

$$\rho(E) = \frac{1}{\pi\sqrt{2}} \frac{r_{\max}}{\sqrt{E}}. \tag{12}$$

Thus, the density of states can be adjusted to meet any requirement since it varies linearly with the box size $r_{\max}$.

An alternative approach, that can yield the continuum wavefunction at any selected energy E employing a fixed basis set, without the need to change box size, is the so-called Galerkin [18], or least-squares, approach [26,274], or free boundary condition approach [245]. It is based on the idea of minimizing the residual vector $R_E = (\mathcal{H} - E)\mathcal{U}$ for a given value of E. Actually, defining

$$b_i = \langle B_i, (\mathcal{H} - E)\mathcal{U}\rangle = \sum_j (\boldsymbol{A}(E))_{ij} c_j \tag{13}$$

where $\mathcal{H}$ is the reduced Hamiltonian as defined in (4) and where

$$(\boldsymbol{A}(E))_{ij} = (\mathcal{H})_{ij} - E(\boldsymbol{S})_{ij}, \qquad \mathcal{U} = \sum_i c_i B_i \tag{14}$$

one minimizes $\sum_i b_i^* b_i$, under the constraint $\sum_i c_i^* c_i = 1$ to avoid the trivial solution $\boldsymbol{c} = 0$. This is equivalent to finding the minimum of

$$\boldsymbol{c}^\dagger \boldsymbol{A}^\dagger \boldsymbol{A} \boldsymbol{c} - \lambda \boldsymbol{c}^\dagger \boldsymbol{c}. \tag{15}$$

The minimum corresponds to the eigenvector of the matrix $\boldsymbol{A}^\dagger \boldsymbol{A}$ whose eigenvalue is closest to zero. Recall that $\boldsymbol{A}^\dagger \boldsymbol{A}$ is Hermitian and has non-negative eigenvalues. Since with a good basis, like B-splines, the lowest eigenvalue is extremely close to zero, and separated from the higher ones by several orders of magnitude, the corresponding eigenvector can be obtained accurately and efficiently employing inverse iteration on the $\boldsymbol{A}^\dagger \boldsymbol{A}$ matrix. Of course this has to be performed for all required values of the energy E. A nice feature of the approach is that it works equally well in the multichannel case. In the latter case, when N_c channels are open, N_c independent solutions are required at each energy E (see section 4.2.1). These solutions are correctly given as N_c independent eigenvectors corresponding to the N_c lowest eigenvalues of $\boldsymbol{A}^\dagger \boldsymbol{A}$. Again, given the large separation from the following eigenvalues, the lowest ones are obtained efficiently by inverse iteration applied to a subspace of dimension N_c. It is important to note two further points.

- Since at a given energy the solution will not be zero at $r_{\max}$, it is necessary to keep the last B-spline, which is nonzero at $r_{\max}$, in the expansion, at variance with the diagonalization approach. It is precisely because of this last function that the B-spline expansion can smoothly follow the solution at each energy, between the energies corresponding to two consecutive knots of the wavefunction at $r_{\max}$, which are given by the diagonalization. The approach is like the solution of the vibrating string problem with only one boundary condition, at the origin, and a free boundary condition at the other end. This allows a solution for any frequency, at variance with the two-point boundary condition, which gives only the eigenfrequencies, and corresponds to the diagonalization process.

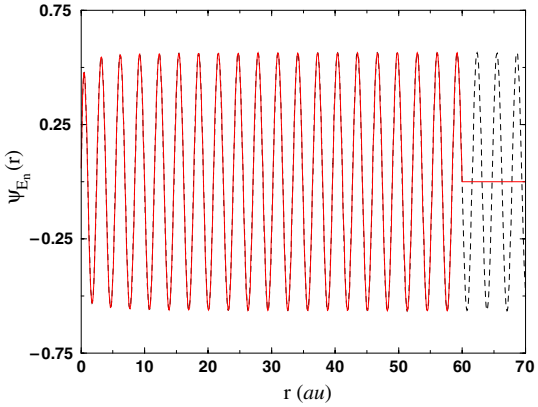


Figure 6. Comparison of a discretized continuum wavefunction (solid curve) with the exact Coulomb wavefunction (dashed curve) in H ($E_k = 54$ eV, $r_{\max} = 60$ au).

(This figure is in colour only in the electronic version)

- Because of the large separation between the eigenvalues corresponding to the required solutions and the rest, the corresponding eigenvectors are very close to those of the $\boldsymbol{A}$ matrix alone, and the same inverse iteration approach can be successfully employed with $\boldsymbol{A}$ [18]. Note that $\boldsymbol{A}$ is non-Hermitian (because of the nonzero spline at $r_{\max}$, in the kinetic energy term), and its eigenvalues are in general complex, so that solutions correspond in this case to minimum-modulus eigenvalues. In practice employing $\boldsymbol{A}$ saves a matrix multiplication, which is expensive; the $\boldsymbol{A}^\dagger\boldsymbol{A}$ formulation is Hermitian, however, and more accurate, and both are currently employed.

Unlike bound states, continuum states are ever-oscillating functions and thus vary up to infinity. Therefore, there are no reasons for not choosing a linear breakpoint sequence which divides the radial coordinates into equally spaced grid points. Although this tends to be the more appropriate sequence for the continuum wavefunctions, in many cases, if not all, one also needs accurate bound states. The idea then is to mix the two types of sequence: one concentrating points near the nucleus, which is better adapted to bound states, and the linear sequence for larger values of r necessary for continuum states (see e.g. expression (206) in appendix A.1).

The maximum energy $E_{\max}$ of the discretized continuum states will be set by both the B-spline order k and the breakpoint spacing, which can be roughly defined as $\Delta r = r_{\max}/N$ (see also the formal discussion of errors at the end of section 2.1). At large distances, a continuum wavefunction oscillates with a wavelength $\lambda = 2\pi/\sqrt{2E}$. The number of B-splines contained in one wavelength has to be large enough to fully reproduce the two sign switches. In practice, one has to achieve $\Delta r = \lambda/3$ for $k = 9$ or $\Delta r = \lambda/5$ for $k = 4$ (these numbers lead to an error on the wavefunction phase that never exceeds 0.1 rad). With such parameters, the typical accuracy for the phase shift is of the order of 10^{-8} rad. This is illustrated in figure 6, where we compare a highly oscillating discretized continuum wavefunction with the exact wavefunction. Note that there is absolutely no difference between the two wavefunctions inside the box.

We now address a final major problem known as the *normalization problem* [343] appearing in any discretization technique. On the one hand, discrete eigenstates are orthonormalized with respect to their (discrete) index as

$$\langle \psi_i | \psi_j \rangle = \delta_{ij}, \tag{16}$$

whereas continuum states, normalized on the (continuous) energy scale, satisfy

$$\langle \psi_E | \psi_{E'} \rangle = \delta(E - E'). \tag{17}$$

On the other hand, the computed discretized continuum states are normalized on the same level as the bound states, i.e. with respect to their index, because they originate from a diagonalization procedure. Any attempt to compute measurable quantities (such as cross-section, ionization rate, ...) involving a continuum state as the final state has to reestablish the correct normalization, that is find the coefficient $\mathcal{A}_n$ such that

$$\psi_{E_n} = \mathcal{A}_n \psi_n \tag{18}$$

where ψ_n is a discretized wavefunction and ψ_{E_n} the corresponding wavefunction normalized on the energy scale. We review here the two methods commonly used with B-splines. The first one compares the discretized state to the exact solution in the asymptotic region ($kr \gg 1$) where the analytical expression is often available (see e.g. equation (9) for the partial waves of a Coulomb potential). A standard fit will provide the normalization factor $\mathcal{A}_{n,l}$ as well as the non-coulombic phase σ_l if the potential is non-coulombic at the origin. This method is actually used in applications involving highly interfering nonperturbative processes [252,253,272] to check the accuracy of the discretized wavefunctions since, apart from the amplitude, it provides the most stringent test on the phase of the wavefunction. Also, it is generally used in the least-squares algorithm when solutions at any energy are obtained [19, 26]. Note that, as shown below, the square of the normalization factor is nothing other than the density of discretized continuum states: $\mathcal{A}_n^2 = \rho(E_n)$. The other technique, which is self-consistent with the discretization procedure, extracts the density of states from the set of eigenenergies without the need for external information. It can be shown [343, 344] that switching from a normalization with respect to the index n to a normalization on the energy scale is achieved by means of the relation

$$\psi_{E_n} = \left| \frac{\partial E(n')}{\partial n'} \right|_{n'=n}^{-\frac{1}{2}} \psi_n, \tag{19}$$

provided $E(n) = E_n$ is a monotonic function of n, which is satisfied in view of the approximate expression (11). Through a Taylor expansion up to third order, one obtains the formal expression

$$\left. \frac{\partial E(n')}{\partial n'} \right|_{n'=n} = \frac{E_{n+1} - E_{n-1}}{2} + \frac{1}{6} \left. \frac{\partial^3 E(n')}{\partial n'^3} \right|_{n'=n}. \tag{20}$$

The absence of the second-order term together with the weak curvature of the function $E(n)$ (see equation (11)) leads to the rather accurate approximation [343]

$$\rho(E_n) = \frac{2}{E_{n+1} - E_{n-1}}. \tag{21}$$

More accurate expressions for the density of states are obtained by polynomial interpolation of E_n and are listed in appendix A.2. Figure 7 clearly shows that the relative error produced by using the approximate expression (21) for evaluating the density of states is already very small. Making use of the five-, seven- or nine-point formula further lowers the error to a level close to machine accuracy with no extra cost, thus reinforcing the use of this self-consistent discretization/normalization procedure.

4. Atomic structure

After the early work by Shore [8–12] and Silverstone and collaborators [15–17] mentioned in the introduction, the first application of B-splines to atomic structure calculations was reported